Practical exercise 2

1. The R vector `abbey` in the package MASS gives 31 determinations of nickel content ($\mu g\ g^{-1}$) in a Canadian syenite rock. It is obtained by:

   ```
   > library(MASS)
   > data(abbey)
   ```

   Produce an Exponential distribution Q-Q plot of these data. Note that there is no prearranged command in R to do this (such as `qqnorm` for the normal distribution), so you will have to use one of the methods described in Question 1 of Computer Practical 1. Hint: one of the commands involves

   ```
   > qexp(ppoints(31))
   ```

   Now find a resistant regression line through the points, passing through the origin. This is obtained by:

   ```
   > lqs(sort(abbey)~qexp(ppoints(31))-1)
   > abline(0,11.84)
   ```

   In the lectures it will be shown that, in fact, the data are probably not well-modelled by an exponential distribution. Investigate instead that some transformation of the data may be modelled by a normal distribution (compare with the methods used for the ozone data).

2. In an experiment at Rothamsted, nocturnal macrolepidoptera (of various species) were caught in a light trap over 4 years (1933-1936). The numbers of insects in a random sample of 15 species were:

   11 54 5 7 4 15 560 18 120 24 3 51 3 12 84

   Enter these results into R and use simple numerical and graphical summaries to investigate the distribution of the data. Note the very considerable positive skewness.

   In addition to the untransformed data, consider the transformations

   (i) $z = \sqrt{y}$
   (ii) $w = \log y$

   Is any of the two transformations of the data likely to be reasonably modelled by a Gaussian (Normal) distribution?

   Also, use a Q-Q plot to investigate whether the untransformed data are reasonably modelled as observations from an exponential distribution.

3. The dataset `chem` in library MASS is made available in R by typing:

   ```
   > library(MASS)
   > data(chem)
   ```

   It contains 24 determinations of copper content ($\mu g\ g^{-1}$) in wholemeal flour. Investigate the distribution of these data. How might the distribution reasonably be described if the one extreme observation were left out?

   The data are believed to come from a larger study which suggests that the mean copper level is $3.68 \mu g\ g^{-1}$. Do you think that the data are compatible with this value of the mean? (Hint: perform an appropriate hypothesis test in R.)